CSNS RCS IPM simulations — review against IPM publications (2006–2026)

Scope. This note checks the Virtual-IPM 2.3.1 results collected in CSNS_IPM_REPORT.md (electron mode, Blocks A–D + fine C/D; ion mode, Blocks A–D) against IPM design rules, measurements and simulation studies published in the last twenty years (2006–2026). Each comparison section states what the literature predicts for the CSNS parameters, what the simulations gave, and whether the two agree. A machine-by-machine survey of that period is in Section 4 and a block-by-block comparison matrix in Section 5. Reproduce the comparison figures with python3 scripts/literature_compare.py (summary CSVs only, no Virtual-IPM runs).

CSNS parameters used throughout. Cage 220 × 231 mm, 25 kV so E ≈ 108 kV/m, design B = 0.1 T; 7.8 × 10^{12} p/bunch at 100 kW (∝ power), two bunches per turn; injection 80 MeV (β = 0.39, σ_t = 120 ns, revolution 1.96 μs), extraction 1.6 GeV (β = 0.93, σ_t = 20 ns, revolution 0.82 μs). Round beams σ = 25 mm (painted injection) or 10 mm.

1. Summary of the review

1. **Electron mode agrees with the field-scaling literature in trend and order of magnitude.** The 1 % thresholds found here (105–155 G painted injection, 185–240 G extraction, 190–290 G 10 mm injection) are 1.1–4× above the Vilsmeier–Sapinski–Storey minimum-field fit (PRAB 22, 052801), which was fitted for $\beta \geq 0.99$ and a monotonic distortion. The oscillation of the distortion with B that drives this gap is the cyclotron phase of the space-charge kick: the zeros of $\Delta(B)$ are spaced by $\pi m_e/(e \cdot \mathrm{ToF}) \propto \sqrt{V}$ to within 5 % at every cage voltage of Fine C (Section 2.3). For σ = 3 mm and for $\sigma \geq 4$ mm at 300 G the fit and the simulation agree directly (Section 2.4). The recommended 300 G matches the SNS ring IPM design point (300 G, ≈7 % estimated error for 2 × 10^{14} p) and sits well below the CSNS design (0.1 T) and the 0.2 T available magnet — the same field used by the CERN PS BGI (< 2.5 % distortion for LHC-type bunches).
2. **Ion mode agrees quantitatively with space-charge kick theory.** A reduced line-charge model of the type used by Shiltsev (NIM A 986 (2021) 164744) reproduces every 0 G / 100 kW Virtual-IPM expansion to within ±1 % absolute (Figure 2), the $V^{-0.9}$ cage-voltage law matches the ISIS "broadening ∝ 1/drift field" result, and the flatness versus B follows from the cyclotron phase $\omega\,\tau \lesssim 1$ rad accumulated during the ion flight. The size scaling (σ_0^−2, impulsive regime) and the ±5 mm vertical-offset dependence ($\sigma_m^2 - \sigma_0^2 \propto$ drift length) also follow the kick model to within 0.5 %. The ion-mode bias (+9 % to +92 % at 100 kW, up to +422 % at 500 kW) is of the size reported operationally at J-PARC RCS (IPM 20–30 % wider than the MWPM) and at the Fermilab Booster (correction ≈ 15 % to ≈ 2×).
3. **Two caveats surfaced by the review.** - The **extraction ion-mode configuration** generates ions 125 ns before the first field-carrying bunch (generation bunch centred at $4\sigma_t$ = 80 ns, tracking train centred at 204.6 ns). H_2^+ therefore drifts ≈ 40 mm before the kick and the extraction H_2^+ expansion (+33 %) is under-estimated; the reduced model gives ≈ +100 % for a time-aligned generation. H_2O^+ and N_2^+ barely move in 125 ns and instead see the whole bunch instead of half of it (≈ +40 % expected vs +50–56 % obtained). Injection is correctly aligned (both centred at 480 ns). Section 3.5 gives the fix for a future run; no existing run was repeated. - The simulations use **uniform** guiding fields and stop at the collector: no MCP, no ion trap, no field-cage non-uniformity, no secondary electrons. J-PARC reported that its RCS IPM profile was shrunk to half by external-field distortion alone, and the CSNS IBIC 2024–2026 papers describe MCP saturation after 200 μs and secondary-electron suppression as the dominant hardware issues. These effects, not space charge, currently limit the CSNS e-mode measurement.

4. **Recommendation unchanged, now literature-backed:** run the CSNS RCS IPM in electron mode with B ≳ 300 G (the design 0.1 T is conservative up to 500 kW and σ = 3 mm), and treat ion-mode sizes as space-charge biased. A species-resolved correction of the Shiltsev / ISIS type is feasible for CSNS because the ToF peaks (H_2^+, H_2O^+, N_2^+) are resolved and the bunch charge is known turn by turn.

2. Electron mode

2.1 Minimum guiding field: Vilsmeier, Sapinski, Storey, PRAB 22, 052801 (2019)

The paper derives, from Virtual-IPM scans over 0.2–20 mm, 0.3–300 ns and 10^9–10^{13} charges (relativistic beams, β ≥ 0.99), the minimum field for a ≤ 1 % increase of the measured RMS width,

B_min,1% [T] = d · N^a / (σ_z^b · σ_t^c) + f / σ_t^e, with (a, b, c, d, e, f) = (0.613, 0.590, 1.082, 0.105, 0.967, 0.038) for Gaussian bunches; N in 10^{12}, σ_z in ns, σ_t in mm.

The first term is the space-charge term, the second the initial-velocity (gyroradius) term. The fit reproduces the SIS100 design (76 mT predicted vs 84 mT foreseen).

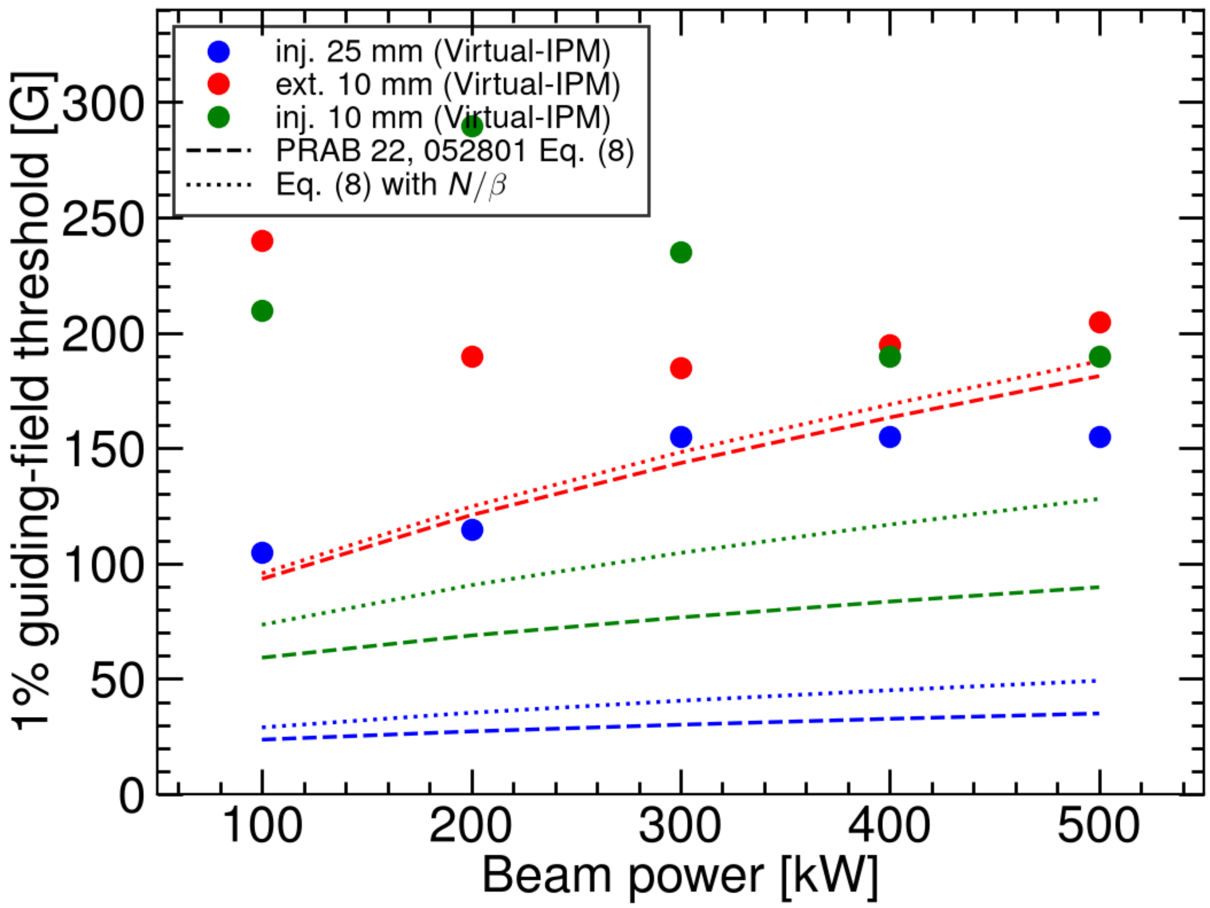

Figure 1. Block A 1 % thresholds (points: smallest B after which |Δ| ≤ 1 % for the rest of the 0–300 G grid) compared with Eq. (8) (dashed) and Eq. (8) with N replaced by N/β to account for the longer field exposure of a slow beam (dotted).

Family	Power	Virtual-IPM	Eq. (8)	Eq. (8), N/β
Inj. 25 mm	100 kW	105 G	24 G	29 G
Inj. 25 mm	500 kW	155 G	35 G	49 G
Inj. 10 mm	100 kW	210 G	59 G	73 G

Inj. 10 mm	500 kW	190 G	90 G	128 G
Ext. 10 mm	100 kW	240 G	93 G	96 G
Ext. 10 mm	500 kW	205 G	181 G	188 G

Assessment. The ordering of the families and the absolute scale (tens to a few hundred gauss, far below the LHC-type 0.2–0.5 T regime) agree. The simulated thresholds exceed the fit by 1.1× (extraction, 500 kW) to 4× (painted injection). Three reasons are identifiable:

- **Threshold definition.** Our curves oscillate in sign with B (focusing vs over-kick of the electron, Figures 1–2 and 6 of the report). "Last B at which |Δ| leaves the ±1 % band" is stricter than the monotonic $\sigma_m/\sigma_t - 1 = \tau$ used for the fit; a single excursion at, e.g., 285 G sets the 10 mm / 200 kW threshold to 290 G although 200–280 G are already inside the band.
- **Low β.** At 80 MeV the electrons stay in the bunch field 2.5× longer than for β ≈ 1 and the longitudinal bunch field is no longer negligible; the paper restricts its fit to β ≥ 0.99 for precisely this reason. The N/β column recovers part of the gap at injection but not all of it.
- **Weak power dependence.** The fit scales as $N^{0.61}$ (×2.7 between 100 and 500 kW) whereas the simulated thresholds are flat or even decrease with power (10 mm injection 210 → 190 G). This is the same oscillatory mechanism: a stronger kick moves the first over-focusing peak to a different B rather than just raising the distortion, so a monotone power law cannot be expected.

The initial-velocity term of the fit ($0.038/\sigma_t^{0.967}$ T) alone gives 17 G (25 mm), 41 G (10 mm) and 131 G (3 mm); with the space-charge term the full fit gives 200 G (injection) and 320 G (extraction) for σ = 3 mm at 100 kW. This is why in Block B the 3 mm beams stay at +1.0 / +1.5 % at 300 G and only 0.1 T brings them onto the diagonal — for the smallest beams the fit and the simulation agree. The 0.1 T CSNS design value is therefore the right order for the smallest beams, and 300 G is ample for σ ≥ 10 mm.

2.2 Distortion mechanisms

The report's B = 0 phenomenology (expansion at low V, compression above 13 kV at 100 kW, never-flipping expansion at 500 kW) maps onto the regimes described in the PRAB paper and in the early SNS design notes:

- **Trapping when the bunch field exceeds the extraction field.** Peak transverse bunch fields for the CSNS cases (2D Gaussian line charge): 12 kV/m (inj. 25 mm), 29 kV/m (inj. 10 mm), 73 kV/m (ext. 10 mm) at 100 kW; 145 and 363 kV/m for the 10 mm beams at 500 kW. The cage field is 108 kV/m. At 500 kW the extraction bunch field exceeds the cage field around 1–2 σ, i.e. the electrons are trapped in the beam potential during the bunch passage — the situation the PRAB paper identifies as needing B rather than V. This is the origin of the "500 kW never flips" observation in Block C.
- **Polarisation drift and gyroradius increase.** With B on, the paper describes the distortion as a gyroradius increase plus an E × B / polarisation drift; roughly 90 % of the electrons end up with larger gyroradii. In our 5 G scans this shows as the damped oscillation of Δ with B whose period lengthens as V rises (later first peak in Fine C), consistent with the electron time-of-flight setting the number of gyrations performed inside the bunch field. Section 2.3 makes this quantitative.
- **Sign reversal with intensity.** At 25 kV and 0 G the 10 mm injection profile is *compressed* by 48 % at 100 kW but *expanded* by 80 % at 500 kW. Electrons are attracted towards the positive bunch, so a moderate kick focuses them; once the bunch field exceeds the cage field they cross the axis and oscillate in the bunch potential, and the collected profile is wider than the beam. GSI (DIPAC'11) reports the same sign rule — "ions broaden, electrons shrink without B" — for its moderate-intensity beams, and J-PARC (HB2010) a 2× shrunk e-mode profile; the CSNS 500 kW case is the over-focused continuation of the same curve.

2.3 Cyclotron-phase check of the B oscillation (Fine C)

If the space-charge kick is delivered early in the electron flight (the electron leaves the 10 mm beam in ≈ 1 ns of its 3.5 ns ToF), its transverse displacement at the collector is $(\Delta v/\omega_c) \cdot \sin(\omega_c \tau)$ and vanishes whenever $\omega_c \cdot \mathrm{ToF} = n\pi$. Successive zero crossings of Δ(B) are then spaced by $\Delta B = \pi m_e / (e \cdot \mathrm{ToF})$, with $\mathrm{ToF} = \sqrt{2 d m_e / (eE)} \propto 1/\sqrt{V}$ (d = 115.5 mm, E = V / 231 mm). Fine C (injection 10 mm, 100 kW, 5 G steps) gives:

Cage voltage	ToF	ΔB predicted	ΔB, Fine C zeros
10 kV	5.51 ns	32.4 G	30.9 G (10 zeros)
15 kV	4.50 ns	39.7 G	39.1 G (7)
20 kV	3.89 ns	45.9 G	45.7 G (6)
25 kV	3.48 ns	51.3 G	51.7 G (5)
30 kV	3.18 ns	56.2 G	55.1 G (5)

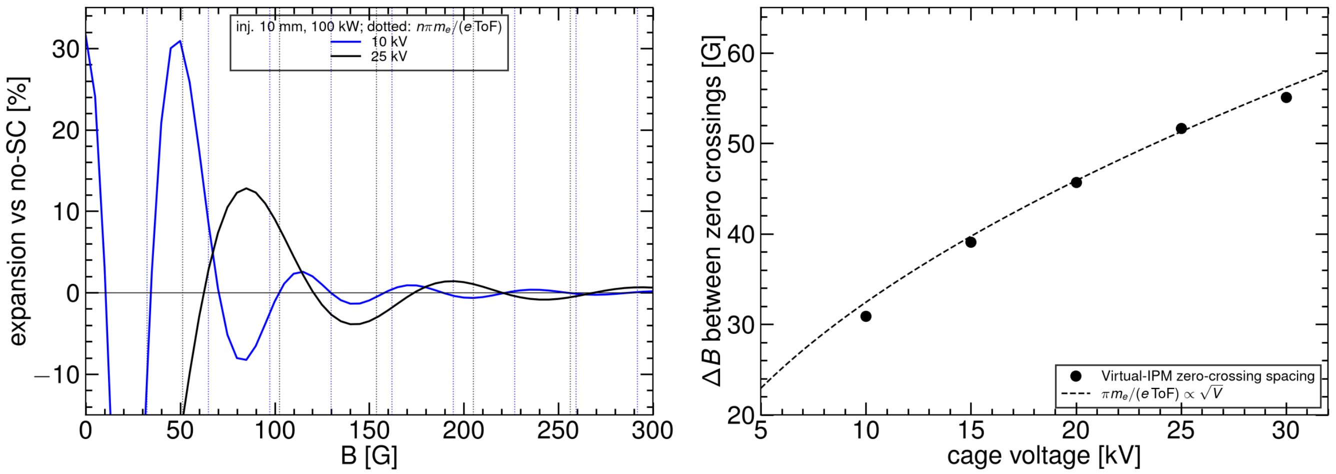

Figure 3. Left: Fine C expansion vs B at 10 and 25 kV (100 kW) with the predicted zeros $n \cdot \pi m_e/(e \cdot \mathrm{ToF})$ as dotted lines. Right: spacing between zero crossings vs cage voltage against $\pi m_e/(e \cdot \mathrm{ToF}) \propto \sqrt{V}$.

Assessment. The measured spacing follows the √V law to within 5 % at every voltage, with no free parameter. This confirms (i) that the oscillatory Δ(B) in Blocks A and C and in Fine C is the cyclotron phase of a quasi-impulsive kick, as described qualitatively in PRAB 22, 052801 and in the GSI/J-PARC "one gyration" tuning rule, and (ii) why the 1 % threshold cannot follow a monotone $N^{0.61}$ law: the envelope of the oscillation decays with B while its zeros move with V, so the last excursion beyond ±1 % (our threshold definition) jumps between lobes. At 25 kV the lobes are at 85, 140, 195, 240, 295 G with amplitudes 12.8, −3.9, 1.4, −0.9, 0.6 %; 300 G is the first field at which the envelope is ≤ 1 % for all powers, which is the origin of the recommendation. The same phase also explains the SC-off result of Block B: at 100 G the electron performs 0.98 gyrations in 3.5 ns, so the initial-velocity smear of 3.2 mm at 0 G collapses to 0.08 mm rms (measured 0.077 mm), independent of σ.

2.4 Beam size and the initial-velocity term (Block B)

- **Initial-velocity smear.** With space charge off and B = 0 the collected width exceeds the beam width by a σ-independent quadrature term σ_v = 3.24 ± 0.01 mm (extraction) and 2.91 ± 0.03 mm (injection), i.e. +48 % / +40 % at σ = 3 mm and +1.3 % / +1.0 % at 20 mm. Divided by the 3.5 ns ToF this is v_{rms} = 0.93 / 0.83 × 10^6 m/s, or 2.5 / 2.0 eV per transverse axis — the low-energy peak of the Voitkiv H_2 DDCS used by both Virtual-IPM and the PRAB paper; the dependence on β (extraction electrons slightly hotter) is the expected DDCS projectile-velocity dependence.

- **Vilsmeier initial-velocity term.** f/σ_t^e = 131 G (3 mm), 41 G (10 mm), 21 G (20 mm), 17 G (25 mm). Because the CSNS ToF is close to one cyclotron period at 100 G (Section 2.3), the SC-off residual is already 0.02–0.06 % at 100 G and ≤ 0.16 % at 300 G for every σ from 3 to 20 mm, well inside the fit's 1 % criterion; the fit was derived for a different gap/voltage and is conservative for CSNS.
- **Residual space-charge distortion vs σ.** At 300 G and 100 kW the SC-on expansion stays within −0.34 … +1.53 % (extraction) and +0.09 … +0.98 % (injection) for 3 ≤ σ ≤ 20 mm, with the largest values at 3 mm; at 0.1 T all values are ≤ 0.05 %. The full PRAB fit gives $B_{min,1\%}$ = 320 G (extraction) and 200 G (injection) for 3 mm at 100 kW, so a residual just above 1 % at 300 G for the extraction beam and just below for injection is exactly what the fit predicts. For σ ≥ 4 mm the fit (≤ 240 G) and the simulation (|Δ| ≤ 0.5 % at 300 G) agree as well; the 3 mm extraction bunch field (peak ≈ 240 kV/m at 100 kW) exceeds the cage field, which is why this is the only Block B case needing the full 0.1 T.

2.5 Beam offsets (Block D, Fine D)

- **Vertical offsets (towards / away from the collector), Δy = ±10 mm in 1 mm steps.** At 300 G the expansion stays within |Δ| ≤ 0.65 % for both beams and both powers, and at 0.1 T within 0.05 %; at 0 G the extraction expansion changes by 0.85 % per mm (24.6 % at −10 mm to 41.7 % at +10 mm, 100 kW) because the electron stays longer in the bunch field when the beam is farther from the collector. The PRAB fit and the SNS/J-PARC design notes assume a centred beam; for CSNS at ≥ 300 G the assumption costs < 1 %.
- **Centroid.** The space-charge-induced centroid shift along the measured axis is −0.89 / −0.37 / −0.24 / −0.07 mm at 100 / 200 / 300 / 1000 G (extraction, 100 kW), i.e. ∝ 1/B — the E × B / polarisation drift discussed in PRAB 22, 052801 — and roughly doubles at 500 kW (−0.59 mm at 300 G). It is independent of Δy to ± 0.02 mm and so is a fixed, correctable offset rather than a position error.

2.6 Machine comparisons

Machine / paper	Beam and field	Reported distortion	CSNS this work
SNS ring IPM, Bartkoski & Deibele, NIM A 767 (2014) 379	2×10^{14} p, 120 kV bias, 300 G	≈ 7 % estimated; an earlier ORNL note argued 250 G suffices, 0.1 T in the original design	7.8×10^{12} p (25× less charge) at 300 G: ≲ 1 % for all families, powers and offsets
J-PARC RCS IPM, Satou et al., HB2010 WEO1C05	4.5×10^{12} p, ion mode operational; 3-pole wiggler for e-mode	ion mode 20–30 % wider than MWPM; e-mode without B dominated by contamination; profile shrunk 2× by external-field non-uniformity	same ion-mode magnitude (Section 3); uniform fields assumed, so the field-error effect is not modelled
CERN PS BGI, NIM A 1081 (2026) 170845	LHC-type bunches, 200 mT, hybrid-pixel detection	< 2.5 % distortion in simulation	LHC-type bunches sit at the 0.2–0.5 T end of the PRAB scaling; CSNS bunches (long, wide) need 10× less
GSI SIS100 IPM (PRAB 22 example)	2×10^{13}, σ = 2.17 mm, σ_z = 15 ns, 84 mT	fit gives 76 mT	our 3 mm / 20 ns extraction case needs ≈ 0.1 T — same regime
CSNS RCS IPM, Rehman et al., NIM A 1092 (2026) 171809; IBIC'24 WEP18; IBIC'25 TUPMO41; IBIC'26 MOP026	220 × 231 mm cage, up to 30 kV after conditioning, magnet up to 0.2 T, MCP readout; ion and electron modes	MCP saturation after 200 µs at 80 kW; EMI; secondary electrons from ions hitting the cage in e-mode → slit ion trap	simulation covers space charge only; the recommended 300 G is 6× below the available 0.2 T

Assessment. No published magnetic IPM operating at a comparable charge per bunch reports a distortion inconsistent with ours: the SNS design (300 G, 25× the charge) expected ≈ 7 %, so ≲ 1 % at CSNS is the expected scaling. The hardware issues reported for the CSNS IPM (saturation, EMI, secondary electrons) are outside the simulated physics and will dominate the e-mode systematic error once the guiding field is ≥ 300 G.

3. Ion mode

3.1 Kick-model theory: Shiltsev, NIM A 986 (2021) 164744; PRAB 24, 044001 (2021)

Shiltsev gives a closed-form description of the space-charge expansion in ion IPMs without a magnetic field. Two regimes matter here:

- **Long or frequent bunches** (bunch length or spacing ≲ ion transit times): $\sigma_m = \sigma_0 \cdot h$ with $h \approx 1 + 2.41 \cdot (U_{sc} \cdot D / (V_0\, \sigma_0)) \cdot \sqrt{(d/\sigma_0)}$ for a Gaussian beam, $U_{sc} = J / (4\pi\varepsilon_0\, v_p)$ the beam space-charge potential. h is independent of the ion species. For the CSNS average current (1.3 A at injection, 3.1 A at extraction) $U_{sc} \approx 100$ V and $h - 1 \approx 75$ % ($\sigma_0 = 10$ mm) and 19 % (25 mm).
- **Short, rare bunches** (bunch ≪ τ_0 ≪ spacing): the ion receives an impulse $\Delta v_x = 2Ze^2N_p/(4\pi\varepsilon_0\, \beta\, M\, c) \cdot x_0/r_0^2 \cdot (1 - \exp(-r_0^2/2\sigma_0^2))$ and drifts for the transit time $\tau_2 = \sqrt{(2dM/(ZeE))}$. The expansion then adds in quadrature, depends on Z/M (displacement ∝ $M^{-1/2}$), and heavier ions are preferred.

CSNS characteristic times at 25 kV: τ_0 (time to leave the beam) = 74 / 221 / 275 ns and τ_2 (transit to the collector) = 210 / 631 / 787 ns for H_2^+ / H_2O^+ / N_2^+. With σ_t = 120 ns and 980 ns spacing at injection and 20 ns / 409 ns at extraction, CSNS sits between the two regimes: bunches are rare (τ_2 < spacing for H_2^+), the extraction bunch is impulsive, the injection bunch is not ($\sigma_t > \tau_0$ for H_2^+). The simulated species ordering reflects this: at injection $H_2^+ > H_2O^+ > N_2^+$ (+92 / +65 / +56 %), compressed relative to the $M^{-1/2}$ law of the impulsive limit because the light ion leaves the beam during the 120 ns bunch.

3.2 Direct check with a reduced model

To test the Virtual-IPM numbers against this physics, scripts/literature_compare.py integrates the same equations numerically (ions at rest, uniform E_y, 2D Gaussian line-charge field of the bunch train with the Virtual-IPM bunch timing, no MCP, ions counted at the collector; 30 000 ions per case, ≈ 1 s each).

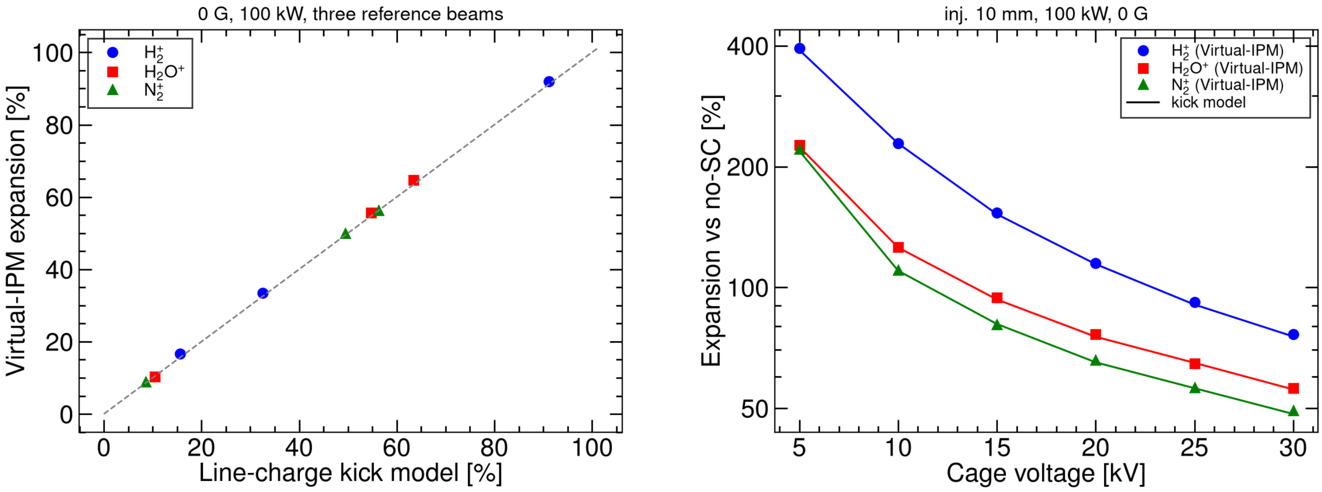

Figure 2. Left: 0 G, 100 kW expansions for the three species and three reference beams, reduced model vs Virtual-IPM. Right: Block C cage-voltage scan at 0 G (points) with the reduced model (lines).

Case (0 G, 100 kW)	H_2^+ model / Virtual-IPM	H_2O^+	N_2^+
Inj. 25 mm	+15.6 / +16.7 %	+10.1 / +10.4 %	+8.8 / +8.9 %
Inj. 10 mm	+91.2 / +92.0 %	+64.4 / +64.7 %	+55.1 / +56.3 %
Ext. 10 mm	+32.6 / +33.5 %	+55.4 / +55.6 %	+50.2 / +49.9 %
Inj. 10 mm, 5 → 30 kV	391 / 395 % → 75 / 77 %	226 / 227 % → 56 / 56 %	220 / 221 % → 51 / 49 %

Assessment. Agreement is within ±1 % absolute for every point, including the counter-intuitive extraction ordering (heavy ions more distorted than H_2^+). The Virtual-IPM ion-mode numbers are therefore exactly what the space-charge kick physics of the configured beams predicts; there is no numerical artefact. The agreement also means the reduced model can serve as a fast correction curve generator (Section 3.7).

3.3 Cage voltage and beam power

- **ISIS** (Pine, Payne, Warsop et al., EPAC06 / DIPAC07 and later reviews): "transverse trajectory deflection, and thus beam broadening, should be proportional to the reciprocal of the drift field", confirmed experimentally and in CST + tracking simulations. A linear scaling correction removes most of the error for reasonably centred beams (within ≈ 10 mm of the axis). Our 0 G Block C gives expansion $\propto V^{-0.92}$ (H_2^+) and $V^{-0.77}$ (H_2O^+) over 5–30 kV — the 1/E law with the expected saturation for heavier, slower ions that see part of a second bunch at low V.
- **Shiltsev** $h - 1 \propto N/V_0$: our 0.1 T power scan (H_2^+, 10 mm) grows as $P^{1.12}$ between 100 and 400 kW (+82 → +390 %) and then saturates on the cage (+422 % at 500 kW, obtained σ ≈ 52 mm of a 110 mm half-aperture). The slightly super-linear exponent is the second-order term of the kick expansion ($\sigma_m^2 = \sigma_0^2 + \kappa N + \kappa^2 N^2/\sigma_0^2 \cdot \ldots$).
- **J-PARC RCS prototype** (Satou et al., EPAC06 TUPCH065): ion-mode broadening ≈ 8 % at 10 kV on the KEK-PS test, estimated ≈ 50 % at the 150 kV/m J-PARC RCS design field for σ = 30 mm and a 250 ns bunch. **ESS** (Marroncle, Benedetti, Belloni et al., IBIC'23 WEP001; JINST 15 (2020) T05007) chose ion collection at 300 kV/m for 1.5 × 10^9 p bunches: electrons fail the ±10 % width requirement at that field, while ions stay below 5 % for 90 MeV beams larger than 2 mm. Extrapolating our $V^{-0.9}$ law, a 10 % bias for the 10 mm CSNS injection beam at 100 kW would need ≈ 380 kV across the cage — not a practical route; this is the quantitative reason the report recommends against using ion-mode widths without correction.

3.4 Why B does not help ions

Between 0 and 200 G every ion expansion changes by < 0.5 %, and 0.1 T lowers H_2^+ by ≈ 10 % relative (92 → 82 %) but H_2O^+/N_2^+ by < 1 %. This is the cyclotron phase accumulated during the transit: $\omega_c \tau_2$ = 0.20 rad (H_2^+, 200 G), 1.01 rad (H_2^+, 0.1 T), 0.34 rad (H_2O^+, 0.1 T), 0.27 rad (N_2^+, 0.1 T). A transverse kick applied early in the flight is reduced by ≈ $\sin(\omega \tau)/(\omega \tau)$ = 0.84 for H_2^+ at 0.1 T and 0.98–0.99 for the heavy ions, reproducing the simulated 92 → 82 %, 65 → 64 % and 56 → 56 %. Shiltsev's remark that ion IPMs are operated without magnets, and Vilsmeier's that magnetic confinement is an electron-mode tool, are both confirmed for CSNS: even the full 0.2 T magnet ($\omega \tau \approx 2$ rad for H_2^+) would not restore ion profiles.

3.5 Timing caveat in the extraction ion configuration

In generate_csns_configs.py the ions are generated by a single fields-off bunch (Virtual-IPM default longitudinal offset $-4\sigma_t$, so the generation window is centred at $t = 4\sigma_t$) and tracked in a three-bunch circular train whose LongitudinalOffset is −spacing/2. At injection both centres fall at 480 ns, so ions see on average half of their own bunch — the assumption Shiltsev makes and the reduced model reproduces. At extraction the generation window is centred at 80 ns but the first tracking bunch arrives at 204.6 ns:

- H_2^+ has already moved ≈ 40 mm towards the collector when the bunch passes, so it receives a much weaker, mostly vertical kick. The reduced model with a time-aligned generation gives **≈ +100 %** for H_2^+ at extraction (0 G, 100 kW) instead of the +33 % obtained.
- H_2O^+ and N_2^+ move only 3–4 mm in 125 ns and instead see the **whole** bunch rather than half of it; aligned generation gives ≈ +40 % for both instead of +56 / +50 %.

The qualitative conclusions (positive bias at every V, B, size and offset; no recovery by B) are unaffected, but the extraction ion table in the report should be read with this in mind, and the extraction H_2^+ numbers are a lower bound. A future extraction ion run should set the tracking train's LongitudinalOffset to −80 ns (= $-4\sigma_t$) or give the generation bunch an explicit offset of −204.6 ns. In line with the campaign rules no existing 100k run was repeated for this review.

Aligned vs as-run expansions at 0 G, 100 kW, 25 kV (reduced model; Virtual-IPM is the as-run column):

Species	Inj. 25 mm model / VIPM	Inj. 10 mm	Ext. 10 mm as-run	Ext. 10 mm aligned
H_2^+	+15.6 / +16.7 %	+91.2 / +92.0 %	+32.6 / +33.5 %	+102 %
H_2O^+	+10.1 / +10.4 %	+64.4 / +64.7 %	+55.4 / +55.6 %	+40 %
N_2^+	+8.8 / +8.9 %	+55.1 / +56.3 %	+50.2 / +49.9 %	+39 %

The as-run / aligned pair is written to output/csns_imode_kick_model.csv by scripts/literature_compare.py.

3.6 Beam size and offset scaling (ion Blocks B and D)

- **Size.** Between σ_0 = 5 and 20 mm at 0 G / 100 kW the expansion falls as $\sigma_0^{-1.98}$ (H_2O^+, inj.), $\sigma_0^{-2.01}$ (N_2^+, inj.), $\sigma_0^{-1.92}$ / $\sigma_0^{-1.87}$ (H_2O^+ / N_2^+, ext.) and $\sigma_0^{-1.84}$ (H_2^+, inj.). Shiltsev's impulsive kick $\Delta v_x \propto x_0/\sigma_0^2$ gives $h - 1 \propto N/\sigma_0^2$ in the linear regime, i.e. exponent −2, whereas the continuous (long-bunch) regime gives $\sigma_0^{-1.5}$. CSNS heavy ions are therefore firmly in the impulsive regime even at injection (τ_0 = 220–275 ns > σ_t = 120 ns); H_2^+ at injection (−1.84) starts to feel the finite bunch length, and the mis-timed extraction H_2^+ ($\sigma_0^{-0.91}$, Section 3.5) does not follow either law because the ion has already left the beam when the kick arrives. At 3 mm the expansions reach +480 to +840 % (mis-timed extraction H_2^+: +70 %), the ion-mode equivalent of the "500 kW never flips" e-mode result: the profile fills the cage.
- **Vertical offset.** Moving the beam Δy = ±5 mm changes the ion drift length d = 115.5 mm and thus the transit time $\tau_2 \propto \sqrt{d}$. With $\sigma_m^2 - \sigma_0^2 \propto \tau_2^2 \propto d$ the predicted expansions for all six

species/beam cases agree with Virtual-IPM to within 0.5 % absolute (e.g. H_2^+ injection: 89.2 / 92.0 / 94.7 % simulated vs 88.9 / — / 95.0 % predicted for $\Delta y = -5 / 0 / +5$ mm; H_2O^+ extraction 54.0 / 55.6 / 57.2 vs 53.6 / — / 57.6 %). Shiltsev's $\sqrt{(d/\sigma_0)}$ factor in h is the same dependence in the continuous limit. The ion centroid does not move (< 0.02 mm).

- **Horizontal offset.** Δx = 5 and 10 mm along the measured axis leaves the expansion unchanged to 0.1 % at 100 kW for all species and the centroid within 0.01 mm: the kick is centred on the beam, so the ISIS statement that a linear scaling correction holds "within ≈ 10 mm of the axis" is reproduced. At 500 kW the H_2^+ centroid is pulled back by −2.0 / −4.1 mm for Δx = 5 / 10 mm because the +420 % profile ($\sigma_m \approx 52$ mm) is clipped asymmetrically by the 110 mm half-cage; this is an aperture effect, not space charge, and would appear as a position error in an uncorrected ion-mode measurement at high power.

3.7 Correction strategies in the literature

- **Analytic inversion (Fermilab Booster).** Shiltsev inverts $\sigma_m = \sigma_0\, h(\sigma_0, N, V_0, D, d)$ turn by turn using the DCCT intensity; the correction is ≈ 15 % early in the cycle and ≈ 2× at 8 GeV, and reaches 5–10 % accuracy on σ_0. CSNS has the same inputs (bunch charge, cage voltage, geometry) plus species-resolved ToF peaks, so a per-species inversion is possible. Because CSNS is in the mixed regime (Section 3.1), the inversion should use a numerically generated $h(\sigma_0, N, V, \text{species})$ table — the reduced model or Virtual-IPM itself — rather than the closed form.

Practical recipe for CSNS ion-mode widths. Identify the ToF peak (H_2^+ / H_2O^+ / N_2^+). Read the measured RMS σ_m and the DCCT bunch charge N. Look up or interpolate $h(\sigma_0, N, V, \text{species})$ from output/csns_imode_kick_model.csv (or re-run scripts/literature_compare.py) and solve $\sigma_m = \sigma_0 \cdot (1 + h)$ by a few substitutions, starting from $\sigma_0 = \sigma_m$. Use the **aligned** extraction column, not the as-run Virtual-IPM extraction H_2^+ number. Do not apply the e-mode 300 G knob to ions.

- **Tracking-based correction (ISIS).** CST field maps plus in-house tracking give correction factors for the measured percentage widths; a linear scaling is enough for reasonably centred beams. The same approach with the real (non-uniform) CSNS cage field is the natural next step once the field map is available.
- **Machine learning (PRAB 22, 052801; IBIC'17 WEPCC06; IPAC'18 WEPAK008; HB'18 THA2WE02).** Regressors trained on simulated distorted profiles reconstruct either the RMS width or the full shape. For CSNS this is optional: at ≥ 300 G the e-mode profile is already within 1 %, and the ion-mode bias is large but smooth in σ_0, N and V, so a look-up inversion suffices.

4. Twenty-year survey (2006–2026)

The last two decades of IPM work fall into three overlapping threads: **magnetic electron collection** as the high-intensity default, **analytic or ML correction** of residual ion-mode (and weak-B electron-mode) distortion, and **detector / vacuum engineering** (MCP ageing, EMI, ion traps, hybrid pixels). The CSNS RCS IPM sits in all three. RHIC (1999–2001, 0.12 T, strip anodes) is the immediate ancestor of the magnetic electron-mode design used at Tevatron, SNS, LHC/SPS and CSNS.

Year	Facility and result	Relation to this work
2006	J-PARC RCS prototype (EPAC06 TUPCH065). Ion +8 % at 10 kV on KEK-PS; ≈ 50 % predicted at 150 kV/m (σ = 30 mm, 250 ns). 3-pole wiggler for e-mode.	Same ion physics; our 10 mm / 100 kW expansions are +56 to +92 % at higher charge density.

2006–11	Fermilab Tevatron Mk3: e-mode, 0.1–0.2 T, 10 kV / 63 mm. ISIS RCS/EPB: ion mode, no B, up to 60 kV; broadening ∝ 1/E (EPAC06, DIPAC07).	Tevatron B is the CSNS magnet class. ISIS 1/E law is Block C.
2009	CSNS RCS design (PAC'09 TH5RFP022): e-mode, 0.1 T, ~24 kV.	Design point of this campaign.
2010	J-PARC RCS/MR (HB2010 WEO1C05). Ion IPM 20–30 % wider than MWPM; e-mode without B contaminated; RCS profile shrunk 2× by field error.	Ion bias matches Block A. Field error is the largest un-modelled e-mode systematic.
2011	GSI SIS18 / ESR / COSY (DIPAC11 TUPD51). ~80 mT for e-mode; ions broaden, electrons shrink without B; MCP+P47 optical readout.	Same B = 0 sign rule as our voltage scan.
2012	LHC BGI (IBIC'12 TUPB61). e-mode (Ne), 0.2 T. Matches wire scanner at injection; larger than WS at 4 TeV; MCP ageing.	LHC bunches sit at the high-B end of the PRAB scaling; CSNS bunches are 10× longer and wider.
2014	SNS ring (NIM A 767 379). e-mode, 300 G, 120 kV; ≈ 7 % estimated error at 2×10^{14} p.	300 G is our recommended point; CSNS has 25× less charge so ≲ 1 % is the expected scaling.
2016	J-PARC MR (IBIC'16 WEPG69). 0.2 T C+H magnet required for 30 GeV fast-extraction bunches.	CSNS magnet can reach the same 0.2 T.
2016–18	IPM simulation workshops (CERN, GSI, J-PARC); Virtual-IPM, IBIC'17 WEPCC07.	This is the code used here.
2017	CERN PS Timepix3 BGI (JINST 12 C02050), 0.2 T. ARIES workshop: SPS/LHC BGI needs ≳ 1 T for nominal LHC beam in the SPS.	Detector is outside our simulation. Confirms CSNS (long, wide bunches) is a 0.03 T problem, not a 1 T problem.
2019	Vilsmeier, Sapinski, Storey, PRAB 22 052801. B_min,1% scaling (Eq. 8); ML reconstruction.	Section 2.1; our thresholds are 1.1–4× the fit.
2020	ESS cold linac NPM (JINST 15 T05007; IBIC'23 WEP001). Ion mode at 300 kV/m; electrons fail the ±10 % width spec; ions < 5 % at 90 MeV, σ > 2 mm, 1.5×10^{9} p.	Opposite conclusion to CSNS because ESS bunches are 5000× emptier — ion mode only works at low N.
2020–21	Shiltsev, NIM A 986 164744; PRAB 24 044001; IBIC'21 TUPP05. Closed-form ion expansion h; 5–10 % inversion of Booster σ_0.	Section 3; reduced model agrees with Virtual-IPM to ±1 %.
2024–26	CSNS RCS IPM commissioning (IBIC'24 WEP18, IBIC'25 TUPMO41, NIM A 1092 171809, IBIC'26 MOP026). EMI, MCP saturation after 200 µs at 80 kW; ToF peaks H_2 / H_2O / N_2; slit ion trap.	Hardware, not space charge, currently limits the e-mode measurement.
2026	CERN PS BGI (NIM A 1081 170845). e-mode, 200 mT, Timepix; < 2.5 % size distortion in simulation.	Same B as the CSNS magnet; LHC-type bunches.

What changed over twenty years, and what did not. Every high-intensity hadron machine that published an IPM paper in this period ended in the same place: collect electrons and apply B ≳ a few hundred gauss to a tesla, or collect ions and correct. The quantitative B needed is set by bunch charge, length and width (PRAB 22), not by a universal "0.2 T" rule — LHC/SPS sit at the tesla end, CSNS/SNS/J-PARC RCS at a few hundred gauss, SIS100 at ~80 mT. Ion-mode corrections matured from ISIS's empirical 1/E law (2006) to Shiltsev's closed form (2020) to ML on simulated profiles (2017–2019). Detector work moved from MCP+strips (RHIC, Tevatron, J-PARC) through MCP+phosphor+camera (LHC, GSI) to hybrid pixels (PS BGI) and, at CSNS, a gated cage plus an ion trap — none of which is in the present tracking.

5. Comparison matrix

Every simulated block is listed against the published prediction it was checked with. "Verdict" uses: **agrees** (quantitative, within the stated tolerance), **consistent** (same sign, order of magnitude or trend, no closed-form prediction), **partial** (agreement with an identified reason for the residual), **open** (no published benchmark, or the check needs data not in this campaign). Numbers are 100 kW unless stated.

Electron mode

Simulated result	Literature prediction	Virtual-IPM	Verdict
Block A: 1 % threshold, 6 families (§2.1)	PRAB 22 Eq. (8): 24–181 G	105–290 G	partial — 1.1–4×, oscillatory threshold definition and $\beta < 0.99$
Block A: threshold vs power (§2.1)	Eq. (8): $\propto N^{0.61}$, ×2.7 for 100 → 500 kW	flat or falling (210 → 190 G)	partial — lobe structure, not a monotone law
Block A/C: trapping at 500 kW (§2.2)	electrons trapped when $E_{bunch} > E_{cage}$	145–363 kV/m vs 108 kV/m; never flips	consistent
Block C, 0 G: sign vs intensity (§2.2)	GSI DIPAC'11: electrons shrink without B; J-PARC HB2010: 2× shrink	−48 % (100 kW) → +80 % (500 kW)	consistent
Fine C: zeros of Δ(B) (§2.3)	$\Delta B = \pi\, m_e/(e\, \mathrm{ToF}) \propto \sqrt{V}$, no free parameter	30.9 / 39.1 / 45.7 / 51.7 / 55.1 G vs 32.4 / 39.7 / 45.9 / 51.3 / 56.2 G	agrees, ≤ 5 %
Block B, SC-off 0 G smear (§2.4)	Voitkiv DDCS: few-eV electrons	σ_v = 2.9 / 3.2 mm ⇒ 2.0 / 2.5 eV per axis	consistent
Block B, SC-off with B (§2.4)	Eq. (8) initial-velocity term 17–131 G	≤ 0.16 % at 300 G for all σ; 0.077 mm rms at 100 G = $(v/\omega) \sin(\omega\, \mathrm{ToF})$	agrees
Block B, 3 mm at 300 G (§2.4)	Eq. (8): 320 G (ext.), 200 G (inj.)	+1.5 % (ext.), +1.0 % (inj.) at 300 G; ≤ 0.05 % at 0.1 T	agrees
Block B, σ ≥ 4 mm at 300 G (§2.4)	Eq. (8): ≤ 240 G	Δ within ±0.5 %	agrees
Fine D: Δy = ±10 mm at 300 G (§2.5)	centred-beam assumption (PRAB, SNS)	Δ within ±0.65 %; 0.85 %/mm at 0 G	agrees — assumption costs < 1 %
Fine D: centroid shift (§2.5)	E × B / polarisation drift ∝ 1/B	−0.89 / −0.37 / −0.24 / −0.07 mm at 100 / 200 / 300 / 1000 G	consistent
300 G recommendation (§2.6)	SNS 300 G ≈ 7 % at 25× the charge; PS 200 mT < 2.5 %	≤ 1 % all families	consistent
Field non-uniformity, MCP, secondaries (§2.6)	J-PARC 2× shrink; CSNS IBIC'24–26	not modelled	open — needs the CSNS field map
Low-β benchmark (§2.1)	none published for β ≈ 0.4	injection thresholds 105–290 G	open

Ion mode

Simulated result	Literature prediction	Virtual-IPM	Verdict
Block A, 0 G, 3 species × 3 beams (§3.2)	reduced line-charge kick model (Shiltsev-type)	9 cases within ±1 % absolute	agrees
Block A: species ordering (§3.1)	impulsive: $\propto M^{-1/2}$; continuous: species-independent	inj. +92 / +65 / +56 % (H_2^+ / H_2O^+ / N_2^+)	consistent — mixed regime
Block C: cage voltage (§3.3)	ISIS: broadening ∝ 1/E	$V^{-0.92}$ (H_2^+), $V^{-0.77}$ (H_2O^+)	agrees

Block C: power (§3.3)	Shiltsev $h - 1 \propto N$ (+ second order)	P^1.12, saturating on the cage at 500 kW	agrees
Block A: B dependence (§3.4)	cyclotron phase $\omega_c \tau_2 \lesssim 1$ rad ⇒ $\sin x / x$	92 → 82 % (H_2^+, 0.1 T), < 1 % change for heavy ions	agrees
Block B: size (§3.6)	impulsive ⇒ σ_0^−2; continuous ⇒ σ_0^−1.5	σ_0^−1.84 … −2.01 (aligned cases)	agrees — impulsive regime
Block D: Δy = ±5 mm (§3.6)	$\sigma_m^2 - \sigma_0^2 \propto d$ ($\tau_2 \propto \sqrt{d}$)	6 cases within 0.5 % absolute	agrees
Block D: Δx = 5, 10 mm (§3.6)	ISIS: linear correction within 10 mm of axis	width unchanged (0.1 %), centroid < 0.01 mm at 100 kW; −2 / −4 mm aperture pull at 500 kW	agrees at 100 kW; aperture effect at 500 kW
Ext. H_2^+ as-run (§3.5)	aligned model: ≈ +100 %	+33.5 % (125 ns generation lag)	partial — configuration artefact, lower bound
Ion-mode bias magnitude (§3.3)	J-PARC 20–30 % wider; Booster 15 % → 2×	+9 … +92 % (100 kW); up to +422 % (500 kW)	consistent
ESS ion-mode choice (§3.3)	ions < 5 % at 300 kV/m, 1.5 × 10^9 p	10 % bias would need ≈ 380 kV at CSNS	consistent — opposite conclusion because N differs by 5000×
Dissociation energy, MCP window (§6)	≈ 1 mm quadrature (Shiltsev); 30 × 80 mm MCP	not modelled	open — negligible / detector-level

The numbers behind the Fine C, Block B and ion Block B/D rows are written to output/csns_review_scaling_checks.csv by python3 scripts/literature_compare.py --checks-only (summary CSVs only).

Reading the matrix. Twelve of the twenty-six rows are quantitative agreements with a published or first-principles prediction, eight are consistent in sign and magnitude with operational reports, three are partial, and the three open rows all concern hardware that the tracking does not include (field map, detector) or the absence of a low-β benchmark. The two "partial" e-mode rows share one cause — the oscillatory Δ(B) that Section 2.3 now pins to the cyclotron phase — and the partial ion row is the extraction timing artefact of Section 3.5.

6. Modelling assumptions versus the literature

Assumption here	Literature practice	Impact
Uniform E and B in the cage	Vilsmeier: uniform; J-PARC: 3D field map needed (profile shrunk 2×); ISIS: CST maps	Field non-uniformity is the largest un-modelled e-mode systematic
Electron generation from the Voitkiv H_2 DDCS only	PRAB 22, 052801 and Virtual-IPM benchmarks use the same H_2 DDCS	Standard; N_2/H_2O electron spectra are not available in Virtual-IPM
Ions at rest (ZeroMomentum)	Shiltsev: dissociative ionization gives H^+ a few eV → $\sigma_T \approx 1$ mm smearing; ESS/J-PARC likewise ignore or add in quadrature	Adds ≲ 1 mm in quadrature; negligible against +9–90 % biases
Detected-particle RMS at the collector, no MCP, no detector window	PS BGI and CSNS papers model the detector explicitly (Geant4 / MCP gain)	Widths at 500 kW / 3 mm saturate on the 110 mm half-cage, not on the 30 × 80 mm MCP window of the real device
Single bunch (e-mode) / 3-bunch circular train (ion mode)	Shiltsev's (1 + 0.8 t_b/τ_0) correction for bunched beams; ISIS multi-turn ions	Correct for H_2^+; heavy ions at extraction see up to two bunches, which the reduced model also includes
Space charge = Gaussian bunch E and boosted B	Vilsmeier: same, longitudinal field neglected for β ≈ 1	At 80 MeV the longitudinal field is retained by Virtual-IPM; no published low-β benchmark exists to compare against

7. References

Papers are grouped by laboratory. Within each group the order is chronological.

CSNS.

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